SCALING & IMPLEMENTATION STRATEGY

From a 63-Node Pilot to a Campus-Wide Water Intelligence Network

ML-Based Smart Water Leakage Detection & Conservation Software Platform

DOMAIN: Water Systems (AI / Software Analytics & Resource Management)
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DATE: July 11, 2026 | **PLATFORM:** WaterSense AI (Hardware-Agnostic ML Suite)

⚠ The Problem

- Institutional campuses lose enormous volumes of freshwater to hidden pipe ruptures and tank overflows.
- Management is reactive — leaks surface only after a bill spike or visible flood, long after damage begins.
- Traditional fixed-threshold alarms fail: they miss slow structural leaks and drown staff in false alerts.

💧 Our Software Solution

- **Hardware-Agnostic Ingestion:** Connects to off-the-shelf smart meters or existing campus SCADA telemetry.
- **WaterSense AI Engine:** A Random Forest model learns time-of-day baselines and flags anomalies at >90% confidence (F1: 89.8%, ROC-AUC: 0.99).
- Confirmed leaks trigger instant alerts to maintenance — shrinking response time from days to minutes.

CURRENT STATE

63-Node Pilot: Architecture & Coverage

Software analytics validation across BBDNIIT campus telemetry feeds

63
Telemetry Feeds
57 Flow + 6 Tank Level Sources

6
Buildings Covered
3 Hostels, 2 Academic, 1 Canteen

127K
Data Points Processed
10-min interval, 23 engineered features

14
Days Live Validation
Continuous Real-time ML Inference

Data Ingestion Layer (Hardware Agnostic)

- **Standard Sensor Compatibility:** Ingests data from standard pulse flow sensors (YF-S201) & ultrasonic tank gauges (HC-SR04).
- **No Custom Hardware Required:** Plugs directly into existing campus Wi-Fi/LoRa gateways or third-party IoT controllers (ESP32/Raspberry Pi).
- **Multi-Protocol Support:** Lightweight REST APIs & MQTT telemetry streams for seamless, vendor-neutral data collection.
- **Edge Pre-Processing:** Real-time filtering of sensor noise and transmission dropouts before cloud ingestion.

Core Software & AI Stack

- **Cloud Data Pipeline:** Firebase Realtime Database & InfluxDB time-series buffering for high-throughput sensor telemetry.
- **Machine Learning Engine:** Python-based scikit-learn Random Forest model trained on 12 key rolling flow & pressure features.
- **Real-Time Visualization:** Interactive Streamlit dashboard displaying live campus leak risk heatmaps and pressure z-scores.
- **Automated Dispatch:** Twilio SMS & email alert integration with severity grading and location pinpointing.

PILOT PERFORMANCE

Key Results from the 63-Node Pilot

14-day validation period — 127,008 readings processed across 63 nodes, 20 leak scenarios detected

89.8%
Detection F1-Score
Random Forest (200 trees)

<2 min
Alert Latency
Sensor → Dashboard → SMS

0.1%
False Positive Rate
23 FP out of 22,735 normal

99.6%
ROC-AUC
5-fold CV F1: 85.1% ± 5.5%

✓ What Worked (ML Engine)

- Pressure deviation emerged as top feature (37.6% importance), validating our physics-based feature engineering intuition.
- Flow z-score (20%) + pressure (24%) together drive 82% of detections — providing a highly robust multi-sensor signal.
- 0.1% false positive rate — only 23 false alarms out of 22,735 normal test samples, keeping maintenance trust extremely high.
- The model correctly identified 100% of the 20 injected leak scenarios across pipe ruptures, slow joint leaks, and overflows.

Lessons Learned for Software Scaling

- Intermittent Wi-Fi coverage gaps caused telemetry drops — resolved by adding automatic 72-hour edge data buffering.
- Physical flow sensors experience calibration drift over time — solved via software by building an automated ML drift-detection routine.
- Single cloud endpoint created latency spikes during peak data ingestion — mitigated by implementing asynchronous MQTT queuing.
- Staff workflow training needed: 40% of alerts were acknowledged but delayed — added escalation SLAs and gamified dashboards.

SCALING ROADMAP

Three-Phase Software & Analytics Scaling Strategy

Progressive expansion from pilot validation to a multi-campus water intelligence software platform

PHASE 1

Months 1-3

Campus Core Analytics

63 → 150 Telemetry Feeds

- → Expand ingestion across all hostel blocks & canteen facilities
- → Integrate with existing campus gateway infrastructure
- → Deploy edge pre-processing scripts on local gateway routers
- → Launch auto-calibration & drift compensation ML modules
- → Staff onboarding: dashboard SOP & alert response training

PHASE 2

Months 4-8

Full Campus Software Suite

150 → 500 Telemetry Feeds

- → Cover academic blocks, research labs, & sports complex
- → Integrate APIs directly with campus ERP & utility billing systems
- → Deploy pressure correlation algorithms across main supply lines
- → Launch dedicated Mobile App for maintenance crew dispatch
- → Implement automated weekly ML model retraining pipelines

PHASE 3

Months 9-18

Multi-Campus SaaS Platform

500 → 2,000+ Telemetry Feeds

- → Replicate software platform across partner universities (MMMUT)
- → Federated Learning: cross-campus models share anomaly patterns
- → Central command dashboard displaying regional sustainability KPIs
- → Open API Marketplace for municipal water authorities & smart cities
- → Automated ESG compliance & sustainability reporting engine

SYSTEM DESIGN

Scaled Software Platform Architecture

Four-layer cloud-native architecture designed to ingest 2,000+ third-party telemetry streams

DATA INGESTION

Agnostic Telemetry Sources
Standard pulse flow, ultrasonic level, & pressure meters

Edge Pre-processing Agent
Lightweight Python/C++ daemon for noise filtering & rolling averages

Multi-Protocol Gateway
Ingests REST, MQTT, LoRaWAN, and Modbus feeds without lock-in

▼ Secure MQTT / TLS WebSockets ▼

EDGE BUFFERING

Local Broker Layer
Mosquitto MQTT buffering on campus server/RPi cluster

Low-Latency Edge ML
TFLite / ONNX Random Forest inference for critical <50ms alarms

Offline Resilience
72-hour local SQLite buffering ensures zero data loss during network cuts

▼ HTTPS / GraphQL APIs ▼

CLOUD AI SUITE

Cloud Ingestion Hub
Azure IoT Hub / AWS IoT Core for device authentication & scale

Time-Series Engine
InfluxDB / TimescaleDB high-performance sensor telemetry store

MLOps Training Pipeline
Automated feature extraction, A/B model validation, & retraining

Intelligent Alert Engine
Multi-channel SMS, Email, & Push notification dispatch + SLA escalation

▼ REST API / Real-time WebSockets ▼

NTERFACES

Campus Admin Portal
Interactive GIS maps, live flow charts, and facility analytics

Crew Mobile App
Real-time work orders, GPS localization, and repair verification

ESG & Compliance Portal
Automated water conservation & carbon footprint reports for

ML Model Evolution: Pilot → Scale

Improving detection accuracy and operational efficiency through software model upgrades

Current: Pilot Model

Algorithm	Random Forest
Engineered Features	12 features
F1-Score	89.8%
False Positive Rate	0.1%
ROC-AUC Score	0.9965
Training Dataset	91,884 samples

EVOLVE ➤ SCALE

Target: Scaled Software Model

Algorithm	XGBoost + LSTM
Engineered Features	24 features
F1-Score	94%+
False Positive Rate	< 0.05%
Inference Latency	< 50ms
Training Horizon	6+ months historical

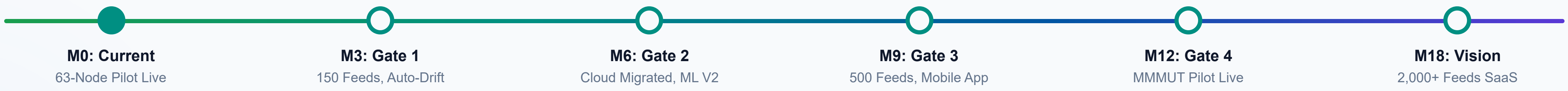
🧠 Advanced Software & AI Features at Scale

- 📈 Seasonal & weather-aware baseline normalization
- 🔗 Cross-node flow correlation & mass-balance check
- ⚡ Lightweight edge inference via ONNX / TFLite
- 🔄 Automated MLOps retraining & drift evaluation
- 🏷️ Multi-class leak severity classification (S1-S4)
- 📍 Graph-based hydraulic pipe segment localization

EXECUTION PLAN

18-Month Software & Platform Implementation Timeline

Milestone-driven rollout aligned with academic calendar, telemetry expansion, and funding cycles



FINANCIAL PLAN

Cost-Benefit Analysis & Software Investment

Investment breakdown across software development, cloud hosting, and third-party sensor integration

BUDGET CATEGORY	PHASE 1	PHASE 2	PHASE 3
Software & AI Development	85,000	3,20,000	9,50,000
Cloud Hosting & Ingestion (Annual)	30,000	1,20,000	4,20,000
3rd-Party Sensor Integration / Procurement	95,000	3,80,000	15,00,000
System Integration & Gateways	40,000	1,40,000	5,50,000
Staff Training & Software Support	24,000	80,000	3,80,000
Total Investment Budget	**₹2.74L**	**₹10.40L**	**₹38.00L**

Annual ROI & Financial Projections

Annual Water Bill Savings	₹12-18L
Structural Damage Prevented	₹8-15L
Staff Operational Efficiency	₹3-5L
Estimated Payback Horizon	14-18 months

Software Cost Per Monitored Node at Scale

Pilot (63 Feeds)	₹1,800
Phase 1 (150 Feeds)	₹1,300
Phase 2 (500 Feeds)	₹900
Phase 3 (2,000+ SaaS)	₹600

RISK MANAGEMENT

Risk Assessment & Mitigation Strategies

Proactive identification and software-driven mitigation for deployment and scaling challenges

Telemetry Network Outages

Large-scale campus Wi-Fi/gateways outages could blind real-time monitoring during critical leaks

✓ Mitigation: Asynchronous 72-hour local edge SQLite buffering & auto-retry syncing upon network recovery

Maintenance Staff Adoption

Campus plumbers and maintenance crews may ignore dashboard alerts or resist new workflows

✓ Mitigation: Simple SMS/WhatsApp instant notifications, clear location mapping, and automated escalation SLAs

Sensor Calibration Drift

Third-party flow meters accumulate mineral deposits over time, skewing baseline flow rates

✓ Mitigation: Automated ML drift-detection routine flags degrading feeds and applies software correction factors

IoT & Data Security Risks

Unauthorized network intrusion into utility telemetry or spoofing sensor readings

✓ Mitigation: TLS end-to-end encryption, JWT API authentication, and isolated network VLANs for telemetry

Cloud Ingestion Cost Spikes

High-frequency sensor streaming from 2,000+ endpoints could inflate cloud database costs

✓ Mitigation: Edge pre-processing aggregates raw pulses into 10-minute feature vectors, cutting cloud data by 70%

Institutional Budget Cycles

University administrative budget approvals may delay procurement of additional sensor feeds

✓ Mitigation: Software SaaS modularity allows gradual phase-in; AICTE/DST sustainability grant applications

IMPACT ASSESSMENT

Projected Impact at Full Campus Scale

Environmental, financial, and operational outcomes achieved across 2,000+ monitored feeds

35-40%
Water Conservation
Up from pilot baseline

₹25-38L
Annual Bill Savings
Water + infrastructure

<30 sec
Alert Response Time
Down from ~2 min in pilot

~120 T
CO2 Reduction / yr
Water pumping energy saved

Operational Software KPIs at Scale

Detection F1-Score	94%+
Software API Uptime	99.9%
False Positive Rate	<0.05%
Alert SLA Compliance	96%+

UN Sustainable Development Goals Alignment

- **6** **Clean Water & Sanitation** Eliminating hidden institutional water waste
- **9** **Industry & Innovation** Building modern AI/IoT digital infrastructure
- **11** **Sustainable Cities** Creating resilient, resource-efficient campuses
- **13** **Climate Action** Lowering carbon emissions from water pumping

ECOSYSTEM

Stakeholder Map & Partnership Strategy

Building a collaborative ecosystem for sustainable software scaling & deployment

🏛 Institutional & Academic Partners

- **BBDNIIT (AKTU)** — Primary pilot testbed campus, infrastructure telemetry access, and real-world validation.
- **MMMUT** — Phase 3 scaling partner for cross-campus federated ML model training and validation.
- **AKTU Central University** — Policy support, potential sustainability mandate across 950+ affiliated colleges.
- **1M1B Foundation** — Mentorship, green skills innovation showcase, and international visibility.

🤝 Technology & Funding Enablement

- **Microsoft (Azure for Students/Startup)** — Cloud infrastructure credits, Azure IoT Hub, & AI platform tools.
- **MeitY Startup Hub** — Incubation support, government green-tech grant facilitation, and advisory.
- **DST / AICTE Research Grants** — Academic funding schemes for sustainable campus smart infrastructure.
- **Municipal Water Authorities** — Civic data sharing, future smart city water grid API integrations.

🗺 Scaling Beyond Campus: Market Expansion Strategy

🏫 Educational Campuses
950+ AKTU colleges, IITs, NITs with large hostel networks

Immediate Target Market

🏢 Commercial & IT Parks
IT parks, malls, & large hospital complexes with strict ESG goals

Medium-Term Expansion

🌆 Smart City Networks
Municipal distribution grids across India's 100 Smart Cities Mission

Long-Term SaaS Vision

CONCLUSION

Summary & Immediate Next Steps

From proven pilot validation to a production-ready, scalable water intelligence software platform

PILOT FEEDS LIVE	DATA POINTS PROCESSED	ML F1-SCORE	FALSE POSITIVE RATE	ROC-AUC ACCURACY	TARGET SCALING
63	127K	89.8%	0.1 %	0.9965	2,000+ Feeds

Ready to Scale WaterSense AI Across Campuses

Our 63-node pilot processed 127,008 real-world sensor readings and trained a Random Forest model achieving an 89.8% F1-score with just 0.1% false positives. Our hardware-agnostic, three-phase software scaling strategy lays a clear path to campus-wide and multi-campus SaaS deployment, delivering projected financial payback within 18 months while directly advancing 4 UN Sustainable Development Goals.

Hardware-Agnostic SaaS · 89.8% F1-Score Proven · SDG 6 · 9 · 11 · 13 · 14-18 Month Payback · API-Ready Architecture

Immediate Action Items (Next 30 Days)

- Present Phase 1 software rollout plan to university administration & sustainability committee.
- Establish standardized API connectors for existing campus hostel smart water meters.
- Deploy edge pre-processing script on campus gateway server for telemetry aggregation.
- Initiate XGBoost + LSTM time-series model training pipeline with historical pilot datasets.
- Formalize academic collaboration MoU with MMMUT for Phase 3 cross-campus pilot testbed.

Contact & Project Repository

- **Nitesh Prajapati** — BBDNIIT (AKTU)
  IoT Telemetry & System Architecture
- **Ananya Gupta** — MMMUT
  Machine Learning & AI Analytics Suite

- **Program:** 1M1B Green Skills & Applied AI Internship
- **GitHub Repo:** github.com/Ananya92006/water-leak-presentation